The Determinant of an Operator and the Virtual K_0-Determinant

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Abstract

The determinant is the scalar action induced by an endomorphism on the top exterior line. The same exterior-line construction governs finite-dimensional linear algebra, algebraic geometry, arithmetic geometry, geometric representation theory, and the virtual K_0-determinant of a Calabi–Yau threefold.

The terminal object is the *virtual K_0-determinant package* on $X = K3 \times E$. The protected K3 elliptic genus $Z_{K3} = 2\phi_{0,1}$ feeds a charge-graded virtual super vector space $\mathcal{V}_{(n,l,m)}$ on Γ_{eff}. Its K_0-determinant in the Borcherds cusp chamber is the Igusa cusp form

$$\mathcal{D}_X(Z) = 64\, q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\text{eff}}} (1 - q^n r^l s^m)^{f(nm,l)} = \Delta_5(Z),$$

a Siegel cusp form of weight 5 on $\mathbf{Sp}_4(\mathbb{Z})$. By Borcherds–Gritsenko–Nikulin, Δ_5 is the denominator of the Borcherds–Kac–Moody superalgebra $\mathfrak{g}_{\Delta_5}$, and

$$\kappa_{\text{BKM}}(\Delta_5) \;=\; \tfrac{1}{2}\, c_1(0) \;=\; 5,$$

the Borcherds weight read off the constant Fourier coefficient $c_1(0) = 10$ of $2\phi_{0,1}$.

The construction supplies the *scalar* virtual K_0-determinant together with its *algebraic target* BKM denominator algebra. It does *not* supply a compact BPS Hilbert space, an operator-level Pfaffian whose square is the BKM denominator squared, or a compact Hall correspondence on $K3 \times E$. The operator-level Pfaffian $\mathrm{Pf}(\mathfrak{D}_X) = \Delta_5$ is a separate construction listed as the open problem F11 in the Vol I frontier. The scalar shadow is constructed here; the operator-level Pfaffian is the distinct second-quantised string construction still to be made. The separation between scalar shadow and operator algebra (scalar-operator) is enforced throughout: every invocation of Δ_5 is labelled with its level (level 5 scalar, not level 4 operator).

Contents

Definition

The determinant of an operator is the scalar by which the operator acts on the top exterior line.

Let V be an n-dimensional vector space over a field k. The top exterior power

$$\det V := \bigwedge^n V$$

is one-dimensional. A linear operator $T : V \to V$ induces a linear operator

$$\bigwedge^n T : \bigwedge^n V \longrightarrow \bigwedge^n V.$$

Every endomorphism of a one-dimensional vector space is multiplication by a unique scalar. That scalar is $\det(T)$:

$$\bigwedge^n T = \det(T) \cdot \mathrm{id}_{\det V}.$$

This is the definition. The formula with permutations, the product of eigenvalues, the Jacobian, the Frobenius determinant in a zeta function, the determinant line on a moduli stack, the Weyl denominator, the central extension of a loop group, and the virtual K_0-determinant of the Mukai stack of $K3$ sheaves whose value on $K3 \times E$ is the Igusa cusp form Δ_5, are realizations of this definition. The last item is the modular-shadow realisation constructed below.

Chapter 1

The Finite-Dimensional Construction

1.1 Alternating Volume Before Determinants

The determinant is forced by one elementary fact: volume is alternating. If two edge vectors of a parallelepiped are equal, the parallelepiped has zero volume. If two edge vectors are exchanged, the orientation changes sign. Algebra encodes this by the exterior power.

Definition 1.1.1 *(alternating multilinear map).* Let V and L be vector spaces over a field k. A map

$$A : V^r \to L$$

is alternating r-linear if it is linear in each variable and

$$A(v_1, \dots, v_r) = 0$$

whenever $v_i = v_j$ for some $i \neq j$.

From this one derives the sign rule. If A is alternating and $i \neq j$, then

$$A(\dots, v_i, \dots, v_j, \dots) = -A(\dots, v_j, \dots, v_i, \dots).$$

Indeed, in the two displayed slots,

$$0 = A(\dots, v_i + v_j, \dots, v_i + v_j, \dots),$$

and multilinearity expands this as the sum of the two swapped terms, because the repeated terms vanish.

Definition 1.1.2 *(exterior power).* The r-th exterior power $\bigwedge^r V$ is the vector space generated by symbols

$$v_1 \wedge \cdots \wedge v_r$$

subject to the relations making the map

$$V^r \to \bigwedge^r V, \qquad (v_1, \dots, v_r) \mapsto v_1 \wedge \cdots \wedge v_r$$

alternating and multilinear.

Equivalently, $\bigwedge^r V$ is the quotient of $V^{\otimes r}$ by the subspace generated by all tensors which express failure of alternation. The quotient formulation proves existence. The symbolic formulation proves usability.

Proposition 1.1.3 *(universal property of exterior powers).* For every vector space L, composition with the wedge map gives a natural bijection

$$\mathrm{Hom}\left(\bigwedge^r V, L\right) \simeq \{\text{alternating } r\text{-linear maps } V^r \to L\}.$$

Proof. A linear map $\phi : \bigwedge^r V \to L$ gives an alternating r-linear map

$$(v_1, \dots, v_r) \mapsto \phi(v_1 \wedge \cdots \wedge v_r).$$

Conversely, if $A : V^r \to L$ is alternating and multilinear, then the defining relations of $\bigwedge^r V$ are exactly the relations that A already satisfies. Hence A descends uniquely to a linear map $\phi_A : \bigwedge^r V \to L$ with

$$\phi_A(v_1 \wedge \cdots \wedge v_r) = A(v_1, \dots, v_r).$$

The two constructions are inverse. □

PROPOSITION 1.1.4 *(basis of an exterior power).* Let V have basis $e_1, \dots, e_n$. Then $\bigwedge^r V$ has basis

$$e_{i_1} \wedge \cdots \wedge e_{i_r}, \qquad 1 \le i_1 < \cdots < i_r \le n.$$

In particular,

$$\dim \bigwedge^r V = \binom{n}{r}.$$

Proof. First, the displayed vectors span. Every wedge

$$v_1 \wedge \cdots \wedge v_r$$

expands multilinearly in the basis e_i. Any term with repeated indices vanishes. Any term with distinct indices can be reordered into increasing order, at the cost of the sign of the permutation.

Second, the displayed vectors are independent. For each increasing multi-index $I = (i_1 < \cdots < i_r)$, define an alternating r-linear form $\varepsilon_I : V^r \to k$ by

$$\varepsilon_I(e_{j_1}, \dots, e_{j_r}) = \begin{cases} \operatorname{sgn}(\sigma), & (j_1, \dots, j_r) = (i_{\sigma(1)}, \dots, i_{\sigma(r)}) \text{ for some } \sigma \in S_r, \\ 0, & \text{otherwise.} \end{cases}$$

By Chapter 1.1.3, ε_I descends to a linear functional on $\bigwedge^r V$. It takes value 1 on $e_{i_1} \wedge \cdots \wedge e_{i_r}$ and value 0 on every other increasing basis candidate. These functionals separate the displayed vectors. Hence the spanning family is independent. □

COROLLARY 1.1.5 *(the top exterior power is a line).* If $\dim V = n$, then

$$\bigwedge^n V$$

is one-dimensional. If $e_1, \dots, e_n$ is a basis of V, then

$$e_1 \wedge \cdots \wedge e_n$$

is a basis of $\bigwedge^n V$.

Proof. Apply Chapter 1.1.4 with $r = n$. There is exactly one increasing n-tuple:

$$(1, 2, \dots, n).$$

□

1.2 Linear Maps Act on Exterior Powers

Let $T : V \to W$ be linear. Define

$$\bigwedge^r T : \bigwedge^r V \to \bigwedge^r W$$

by

$$(\bigwedge^r T)(v_1 \wedge \cdots \wedge v_r) := T(v_1) \wedge \cdots \wedge T(v_r).$$

This is well-defined because the right side is alternating and multilinear in $v_1, \ldots, v_r$. The universal property of the exterior power is exactly the statement that this prescription descends to a linear map.

Proposition 1.2.1 *(functoriality).* If $V \xrightarrow{T} W \xrightarrow{S} U$ are linear maps, then

$$\bigwedge^r (S \circ T) = (\bigwedge^r S) \circ (\bigwedge^r T).$$

Also

$$\bigwedge^r (\mathrm{id}_V) = \mathrm{id}_{\bigwedge^r V}.$$

Proof. Both identities are checked on wedge generators. For example,

$$\bigwedge^r (S \circ T)(v_1 \wedge \cdots \wedge v_r) = S(Tv_1) \wedge \cdots \wedge S(Tv_r),$$

which is exactly

$$(\bigwedge^r S)(Tv_1 \wedge \cdots \wedge Tv_r) = ((\bigwedge^r S) \circ (\bigwedge^r T))(v_1 \wedge \cdots \wedge v_r).$$

□

1.3 Definition of the Determinant

Specialize to $W = V$, $r = n = \dim V$, and $T : V \to V$. Functoriality gives

$$\bigwedge^n T : \bigwedge^n V \to \bigwedge^n V.$$

By Chapter 1.1.5, the source and target form a one-dimensional line. Hence the operator is multiplication by one scalar.

Definition 1.3.1 *(determinant of an operator).* Let V be finite-dimensional and $T : V \to V$ linear. The determinant $\det(T) \in k$ is the unique scalar satisfying

$$\bigwedge^{\dim V} T = \det(T) \cdot \mathrm{id}_{\bigwedge^{\dim V} V}.$$

The rest of finite-dimensional determinant theory evaluates this scalar in useful coordinates.

1.4 One Dimension

Let V be one-dimensional. Choose a non-zero vector e. Every linear operator $T : V \to V$ has the form

$$T(e) = ae$$

for a unique scalar $a \in k$. In one dimension the determinant is this scalar:

$$\det(T) = a.$$

This case already contains the whole idea. A one-dimensional vector space is a line. An operator on a line is multiplication by a scalar. The determinant asks every operator, even an operator on a large space, to act on a line canonically attached to that space.

1.5 Two Dimensions by Area

Let V have basis e_1, e_2. Write

$$T(e_1) = ae_1 + ce_2, \qquad T(e_2) = be_1 + de_2.$$

The matrix of T in the basis (e_1, e_2) is

$$[T] = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

The oriented area element is $e_1 \wedge e_2$. The wedge product is bilinear and alternating:

$$u \wedge v = -v \wedge u, \qquad u \wedge u = 0.$$

Therefore

$$\begin{aligned} T(e_1) \wedge T(e_2) &= (ae_1 + ce_2) \wedge (be_1 + de_2) \\ &= ab\, e_1 \wedge e_1 + ad\, e_1 \wedge e_2 + cb\, e_2 \wedge e_1 + cd\, e_2 \wedge e_2 \\ &= ad\, e_1 \wedge e_2 - bc\, e_1 \wedge e_2 \\ &= (ad - bc) e_1 \wedge e_2. \end{aligned}$$

Hence

$$\det(T) = ad - bc.$$

Nothing was memorized. The formula is forced by the action on the oriented area line $\bigwedge^2 V$.

1.6 Three Dimensions by Volume

Let V have basis e_1, e_2, e_3. The oriented volume line is

$$\det V = \bigwedge^3 V = k \cdot (e_1 \wedge e_2 \wedge e_3).$$

If

$$T(e_j) = \sum_{i=1}^{3} a_{ij} e_i,$$

then

$$T(e_1) \wedge T(e_2) \wedge T(e_3) = \left(\sum_{\sigma \in S_3} \operatorname{sgn}(\sigma) a_{\sigma(1),1} a_{\sigma(2),2} a_{\sigma(3),3} \right) e_1 \wedge e_2 \wedge e_3.$$

Terms with repeated basis vectors vanish because the wedge product is alternating. Terms using each basis vector once survive. The sign is the sign of the permutation needed to reorder the surviving wedge into $e_1 \wedge e_2 \wedge e_3$.

Thus the determinant is the signed sum over all ways of selecting one entry from each column and one entry from each row.

1.7 The General Formula

Proposition 1.7.1 *(permutation formula).* Let V have basis $e_1, \ldots, e_n$. Let $T(e_j) = \sum_i a_{ij} e_i$. Then

$$T(e_1) \wedge \cdots \wedge T(e_n) = \det(a_{ij})\, e_1 \wedge \cdots \wedge e_n,$$

where

$$\det(a_{ij}) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1),1} \cdots a_{\sigma(n),n}.$$

Proof. Expand the wedge product by multilinearity:

$$\left(\sum_{i_1} a_{i_1,1} e_{i_1} \right) \wedge \cdots \wedge \left(\sum_{i_n} a_{i_n,n} e_{i_n} \right).$$

If two indices i_r, i_s coincide, the term contains the same basis vector twice and vanishes. The only surviving terms are indexed by permutations $\sigma \in S_n$, with $i_j = \sigma(j)$. For such a term,

$$e_{\sigma(1)} \wedge \cdots \wedge e_{\sigma(n)} = \operatorname{sgn}(\sigma)\, e_1 \wedge \cdots \wedge e_n.$$

Substitution gives the displayed formula. □

1.8 Basis Independence

The permutation formula depends on a basis. The determinant is basis-free.

Proposition 1.8.1 *(basis-free determinant).* Let V be finite-dimensional. There is a unique scalar $\det(T)$ such that

$$\bigwedge^{\dim V} T = \det(T) \cdot \mathrm{id}_{\det V}.$$

This scalar is independent of every choice of basis.

Proof. The line $\det V = \bigwedge^n V$ is one-dimensional. Hence every endomorphism of $\det V$ is multiplication by a unique scalar. The induced map $\bigwedge^n T$ is such an endomorphism. The scalar is defined without choosing a basis. If a basis is chosen, the scalar is computed by the permutation formula of Chapter 1.7; changing basis changes both the chosen generator of $\det V$ and its image by the same non-zero scalar, so the multiplier is unchanged. □

1.9 What the Determinant Measures

Proposition 1.9.1 *(detecting invertibility).* For $T : V \to V$ with V finite-dimensional,

$$\det(T) \neq 0 \iff T \text{ is invertible.}$$

Proof. If T is singular, its image has dimension $< n$. Then $T(e_1), \ldots, T(e_n)$ are linearly dependent for every basis (e_i). The wedge of linearly dependent vectors is zero, so $\bigwedge^n T = 0$, hence $\det(T) = 0$.

If T is invertible, $T(e_1), \ldots, T(e_n)$ is a basis. Its wedge is a non-zero generator of $\det V$. Hence $\bigwedge^n T$ is non-zero on a one-dimensional space, so it is multiplication by a non-zero scalar. □

Proposition 1.9.2 *(multiplicativity).* For $S, T : V \to V$,

$$\det(ST) = \det(S)\det(T).$$

Proof. Exterior powers are functorial:

$$\bigwedge^n(ST) = (\bigwedge^n S)(\bigwedge^n T).$$

On the determinant line, these maps are multiplication by $\det(S)$ and $\det(T)$. Their composite is multiplication by $\det(S)\det(T)$. □

Proposition 1.9.3 *(eigenvalue product).* If T has eigenvalues $\lambda_1, \ldots, \lambda_n$ counted with algebraic multiplicity over an algebraic closure of k, then

$$\det(T) = \lambda_1 \cdots \lambda_n.$$

Proof. Over an algebraic closure, triangularize T. In an upper triangular matrix, the determinant formula has only one non-zero surviving diagonal contribution:

$$\det(T) = a_{11} \cdots a_{nn}.$$

The diagonal entries of a triangular form are the eigenvalues counted with algebraic multiplicity. The determinant is unchanged by extension of scalars. □

1.10 Worked Linear Examples

1.10.1 Diagonal Scaling

Let $V = k^3$ with basis e_1, e_2, e_3, and let

$$T(e_1) = ae_1, \qquad T(e_2) = be_2, \qquad T(e_3) = ce_3.$$

Then

$$T(e_1) \wedge T(e_2) \wedge T(e_3) = (ae_1) \wedge (be_2) \wedge (ce_3) = abc\, e_1 \wedge e_2 \wedge e_3.$$

Thus

$$\det(T) = abc.$$

The determinant is the product of the three independent stretch factors.

1.10.2 *Shear*

Let $V = k^2$, and let

$$T(e_1) = e_1, \qquad T(e_2) = e_2 + ae_1.$$

Then

$$T(e_1) \wedge T(e_2) = e_1 \wedge (e_2 + ae_1) = e_1 \wedge e_2 + a\, e_1 \wedge e_1 = e_1 \wedge e_2.$$

Therefore

$$\det(T) = 1.$$

A shear changes shape and preserves oriented area.

1.10.3 *Projection*

Let $V = k^2$, and let

$$P(e_1) = e_1, \qquad P(e_2) = 0.$$

Then

$$P(e_1) \wedge P(e_2) = e_1 \wedge 0 = 0.$$

Hence

$$\det(P) = 0.$$

Projection collapses area to a line. The determinant vanishes because the top exterior image vanishes.

1.10.4 *Rotation over* $\mathbb{R}$

Let $V = \mathbb{R}^2$, and let

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Then

$$R_\theta(e_1) = \cos\theta\, e_1 + \sin\theta\, e_2, \qquad R_\theta(e_2) = -\sin\theta\, e_1 + \cos\theta\, e_2.$$

Therefore

$$\begin{aligned} R_\theta(e_1) \wedge R_\theta(e_2) &= (\cos\theta\, e_1 + \sin\theta\, e_2) \wedge (-\sin\theta\, e_1 + \cos\theta\, e_2) \\ &= \cos^2\theta\, e_1 \wedge e_2 + \sin^2\theta\, e_1 \wedge e_2 \\ &= e_1 \wedge e_2. \end{aligned}$$

Thus

$$\det(R_\theta) = 1.$$

Rotation preserves oriented area.

1.11 Exact Sequences and Block Operators

The determinant line behaves well in exact sequences. This is the first point at which the finite-dimensional definition becomes a machine for geometry.

Proposition 1.11.1 *(determinant line of a short exact sequence).* Let

$$0 \longrightarrow U \longrightarrow V \longrightarrow W \longrightarrow 0$$

be a short exact sequence of finite-dimensional vector spaces. There is a canonical isomorphism

$$\det V \simeq \det U \otimes \det W.$$

Proof. Choose a basis $u_1, \dots, u_r$ of U. Choose vectors $\widetilde{w}_1, \dots, \widetilde{w}_s \in V$ lifting a basis $w_1, \dots, w_s$ of W. Then

$$u_1 \wedge \cdots \wedge u_r \wedge \widetilde{w}_1 \wedge \cdots \wedge \widetilde{w}_s$$

is a basis of $\det V$. Send it to

$$(u_1 \wedge \cdots \wedge u_r) \otimes (w_1 \wedge \cdots \wedge w_s).$$

Changing the lifts $\widetilde{w}_i$ adds U-vectors to them; the extra terms contain repeated U-directions in the top wedge and vanish. Changing bases multiplies both sides by the same determinant. Hence the map is canonical. □

Corollary 1.11.2 *(block triangular determinant).* If $T : V \to V$ preserves a subspace $U \subset V$, and if $\overline{T}$ is the induced operator on V/U, then

$$\det(T) = \det(T|_U)\det(\overline{T}).$$

Proof. Apply det to the exact sequence

$$0 \to U \to V \to V/U \to 0$$

and use Chapter 1.11.1. On the line $\det U \otimes \det(V/U)$, the induced action is multiplication by $\det(T|_U)\det(\overline{T})$. □

Chapter 2

From Operators to Determinant Lines

2.1 The Determinant of a Vector Bundle

Let X be a scheme or a smooth manifold. A rank n vector bundle E assigns an n-dimensional vector space E_x to every point $x \in X$, varying algebraically or smoothly. Its determinant is the line bundle

$$\det E := \bigwedge^n E.$$

Fibrewise,

$$(\det E)_x = \det(E_x).$$

Thus the finite-dimensional determinant line globalizes without change. A vector bundle is a family of vector spaces. Its determinant line is the family of their top exterior lines.

2.2 Determinant of a Bundle Endomorphism

Let $T : E \to E$ be an endomorphism of a rank n vector bundle. Taking top exterior powers gives

$$\bigwedge^n T : \det E \to \det E.$$

An endomorphism of a line bundle is a section of $\mathcal{O}_X$. Thus

$$\det(T) \in \Gamma(X, \mathcal{O}_X).$$

At a point x,

$$\det(T)(x) = \det(T_x : E_x \to E_x).$$

The zero locus of this function is exactly the singular locus of T_x.

2.3 The Determinant of a Complex

Geometry rarely gives a single vector space. It gives cohomology in several degrees. The determinant line of a complex is first a chain-level alternating product; cohomology gives a canonically isomorphic line by the Knudsen–Mumford determinant isomorphism.

Definition 2.3.1 *(determinant line of a finite complex).* Let $C^\bullet$ be a bounded complex of finite-dimensional vector spaces. Define the chain-level determinant

$$\det_{\mathrm{KM}} C^\bullet := \bigotimes_i \left(\det C^i\right)^{(-1)^i},$$

where $L^{+1} = L$ and $L^{-1} = L^\vee$. There is a canonical Knudsen–Mumford isomorphism

$$\det_{\mathrm{KM}} C^\bullet \simeq \bigotimes_i \left(\det H^i(C^\bullet)\right)^{(-1)^i}.$$

The inverse in odd degree is load-bearing. It is the line-level form of the alternating sign in Euler characteristics:

$$\chi(C^\bullet) = \sum_i (-1)^i \dim H^i(C^\bullet).$$

The determinant line is the multiplicative Euler characteristic.

Example 2.3.2 *(two-term complex).* Let

$$C^\bullet = [C^0 \xrightarrow{d} C^1].$$

Then

$$\det_{\mathrm{KM}} C^\bullet = \det C^0 \otimes \det C^{1\vee} \simeq \det H^0(C^\bullet) \otimes \det H^1(C^\bullet)^\vee.$$

If d is an isomorphism, both cohomology groups vanish and $\det_{\mathrm{KM}} C^\bullet \simeq k$. Acyclic complexes have a differential trivialisation. Torsion is the comparison between that trivialisation and a chosen basis or metric trivialisation.

2.4 Why the Signs Are Forced

Suppose $C^\bullet$ is a complex with zero differential:

$$C^\bullet = \cdots \to 0 \to C^0 \to C^1 \to 0 \to \cdots.$$

Then $H^i(C^\bullet) = C^i$, so

$$\det_{\mathrm{KM}} C^\bullet = \det C^0 \otimes (\det C^1)^\vee.$$

If one thinks of C^0 as numerator and C^1 as denominator, then the determinant line remembers the cancellation encoded by the differential. This is the same alternating structure as

$$\det(1 - tT|C^\bullet) := \prod_i \det(1 - tT|C^i)^{(-1)^i}.$$

The signs are the price of making cohomology invariant under homotopy.

Chapter 3

Infinite-Dimensional Operators

3.1 The Obstruction

The finite-dimensional determinant used the line $\bigwedge^n V$. If V is infinite-dimensional, there is no final exterior power. There is no largest n. Thus there is no automatic line $\det V$, and no automatic scalar $\det(T)$.

Every infinite-dimensional determinant is therefore extra structure. One must specify:

(i) a class of operators for which a determinant is defined;
(ii) a replacement for the finite product of eigenvalues;
(iii) a topology or regularisation making the replacement converge;
(iv) functorial identities preserved by the construction.

3.2 Fredholm Determinants

Let $\mathcal{H}$ be a separable Hilbert space and let $K : \mathcal{H} \to \mathcal{H}$ be trace-class. The operator $1 + K$ has a Fredholm determinant

$$\det(1 + K).$$

If K has finite rank with eigenvalues $\lambda_1, \dots, \lambda_m$ on its non-zero image, then

$$\det(1 + K) = \prod_{j=1}^{m} (1 + \lambda_j).$$

For trace-class K, the infinite product over eigenvalues is convergent after the trace-class hypothesis is imposed:

$$\det(1 + K) = \prod_{j} (1 + \lambda_j).$$

Equivalently,

$$\det(1 + K) = \sum_{r \geq 0} \operatorname{Tr}\left(\bigwedge^{r} K \right),$$

where the series converges for trace-class K. This formula is the finite-dimensional exterior-power construction with all exterior degrees retained, because there is no top degree.

3.3 Zeta-Regularised Determinants

Let A be a positive self-adjoint operator with discrete positive spectrum

$$0 < \lambda_1 \leq \lambda_2 \leq \cdots$$

and finite multiplicities. Suppose the spectral zeta function

$$\zeta_A(s) := \sum_j \lambda_j^{-s}$$

converges for $\mathrm{Re}(s) \gg 0$ and admits meromorphic continuation to a neighbourhood of $s = 0$, regular at 0. Define

$$\det_\zeta(A) := \exp\big(-\zeta_A'(0)\big).$$

This definition is forced by the finite case. If the spectrum is finite, then

$$\zeta_A'(s) = \sum_j -(\log \lambda_j)\lambda_j^{-s},$$

so

$$\zeta_A'(0) = -\sum_j \log \lambda_j.$$

Therefore

$$\exp(-\zeta_A'(0)) = \exp\left(\sum_j \log \lambda_j\right) = \prod_j \lambda_j.$$

The zeta determinant is the finite product of eigenvalues continued through the spectral zeta function.

3.4 THE DETERMINANT OF $s - \Theta$

In arithmetic applications one often wants a determinant depending on a complex parameter s:

$$\det_\infty(s\,\mathrm{id} - \Theta).$$

If Θ has discrete spectrum $\{\rho\}$, this determinant is the regularised form of

$$\prod_\rho (s - \rho).$$

This product diverges in the arithmetic cases of interest. A regularised determinant replaces it by a function with the required logarithmic derivative:

$$\frac{d}{ds} \log \det_\infty(s\,\mathrm{id} - \Theta) = \mathrm{Tr}_{\mathrm{reg}}\big((s\,\mathrm{id} - \Theta)^{-1}\big).$$

Thus a determinant packages a spectrum into a function. Conversely, if a function is known by its zeros, a determinant formula asserts that those zeros are the spectrum of an operator.

This is why determinant formulas in arithmetic are severe. Writing

$$\xi(s) = \det_\infty(s\,\mathrm{id} - \Theta)$$

strengthens the zero-set statement for ξ: the zeros are realised by an operator Θ on a cohomological space, with the determinant regularisation compatible with trace formulas.

Chapter 4

Algebraic Geometry

4.1 The Jacobian as a Determinant

Let $f : X \to Y$ be a morphism of smooth n-dimensional varieties over a field k. The differential is a bundle map

$$df : f^* \Omega^1_Y \to \Omega^1_X.$$

Taking determinants gives

$$\det(df) : f^* \det \Omega^1_Y \longrightarrow \det \Omega^1_X.$$

Locally, if Y has coordinates $y_1, \ldots, y_n$ and X has coordinates $x_1, \ldots, x_n$, then

$$df(dy_j) = \sum_i \frac{\partial f_j}{\partial x_i} dx_i.$$

Therefore

$$df(dy_1) \wedge \cdots \wedge df(dy_n) = \det\left(\frac{\partial f_j}{\partial x_i}\right) dx_1 \wedge \cdots \wedge dx_n.$$

The classical Jacobian determinant is the determinant of the differential on the canonical line.

4.2 Canonical Bundle

For a smooth n-dimensional variety X, the canonical bundle is

$$\omega_X := \det \Omega^1_X = \bigwedge^n \Omega^1_X.$$

This is the line of algebraic volume forms. The determinant construction is the reason the canonical bundle is a line: top forms are top exterior powers of cotangent spaces.

If $f : X \to Y$ is étale of relative dimension 0, then df is an isomorphism. Hence $\det(df)$ is non-zero everywhere. Conversely, for a finite separable map between smooth curves, vanishing of $\det(df)$ detects ramification. In dimension 1, this is the ordinary derivative. In higher dimension, it is the determinant of the Jacobian matrix.

4.3 Chern Classes and the Determinant

For a vector bundle E of rank n, the first Chern class is the first Chern class of its determinant line:

$$c_1(E) = c_1(\det E).$$

The determinant extracts the line-bundle shadow of E. Many intersection-theoretic invariants of E begin with this shadow.

If $E = L_1 \oplus \cdots \oplus L_n$ splits as line bundles, then

$$\det E = L_1 \otimes \cdots \otimes L_n,$$

so

$$c_1(E) = c_1(L_1) + \cdots + c_1(L_n).$$

This is the line-bundle analogue of the eigenvalue product formula.

4.4 Determinant of Cohomology

Let X be a proper scheme over k, and let E be a coherent sheaf whose cohomology groups are finite-dimensional. Define

$$\lambda(E) := \det R\Gamma(X, E) = \bigotimes_i \det H^i(X, E)^{(-1)^i}.$$

This line is the determinant of cohomology.

For a smooth projective curve C,

$$\lambda(E) = \det H^0(C, E) \otimes \det H^1(C, E)^\vee.$$

If $\pi : X \to S$ is proper perfect and E is an S-perfect complex, the determinant of cohomology is the line bundle

$$\det R\pi_* E \in \mathrm{Pic}(S).$$

Its fibre over $s \in S$ identifies with $\det R\Gamma(X_s, E_s)$ under the usual cohomology-and-base-change hypotheses; the individual cohomology dimensions may jump. Theta line bundles on moduli spaces are determinant-of-cohomology line bundles in this perfect relative sense.

4.5 Degeneracy Loci

Let $\phi : E \to F$ be a morphism of vector bundles of equal rank n on X. Taking determinants gives

$$\det(\phi) : \det E \to \det F.$$

Equivalently,

$$\det(\phi) \in \Gamma(X, \det F \otimes (\det E)^\vee).$$

The vanishing locus of this section is the rank-drop locus of ϕ_x. Thus a determinant turns a rank condition into a divisor when the expected codimension is 1.

More general degeneracy loci use minors. A minor is the determinant of a map between exterior powers:

$$\bigwedge^r E \to \bigwedge^r F.$$

Thus even the higher rank stratification is built from determinant maps.

4.6 Derived Algebraic Geometry

In derived algebraic geometry the basic objects are perfect complexes. The determinant extends to a functor

$$\det : \mathrm{Perf}(X) \to \mathrm{Pic}(X),$$

where $\mathrm{Pic}(X)$ is the Picard groupoid of line bundles. Locally, if a perfect complex is represented by vector bundles E^i, then

$$\det(E^\bullet) = \bigotimes_i (\det E^i)^{(-1)^i}.$$

Passing to cohomology gives the same line when the complex has finite cohomology. This determinant functor is the line-level shadow of the derived category.

Algebraic geometry uses determinants because geometry produces families. A determinant is the functor which turns a family of finite-dimensional linear problems into a line bundle, and line bundles have divisors, Chern classes, metrics, and moduli.

Chapter 5

Arithmetic Geometry

5.1 The Fundamental Trace-to-Determinant Identity

Proposition 5.1.1 *(trace–determinant identity).* Let T be an endomorphism of a finite-dimensional vector space. Then

$$-\log\det(1-tT)=\sum_{r\geq 1}\frac{\mathrm{Tr}(T^r)}{r}t^r$$

as a formal power series.

Proof. After extending scalars, triangularize T, with eigenvalues $\lambda_1,\dots,\lambda_n$. Then

$$\det(1-tT)=\prod_j(1-t\lambda_j).$$

Hence

$$-\log\det(1-tT)=-\sum_j\log(1-t\lambda_j)=\sum_j\sum_{r\geq 1}\frac{(t\lambda_j)^r}{r}.$$

Interchange the finite sum over j with the formal sum over r:

$$\sum_{r\geq 1}\frac{t^r}{r}\sum_j\lambda_j^r.$$

But $\sum_j\lambda_j^r=\mathrm{Tr}(T^r)$. This proves the identity. □

This identity is the bridge from point counts to zeta functions.

5.2 Varieties over Finite Fields

Let $X/\mathbb{F}_q$ be a separated variety of finite type. Its zeta function is

$$Z(X,t):=\exp\left(\sum_{r\geq 1}\#X(\mathbb{F}_{q^r})\frac{t^r}{r}\right).$$

For $\ell\neq p$ and geometric Frobenius Frob_q, the Grothendieck–Lefschetz trace formula uses compactly supported étale cohomology:

$$\#X(\mathbb{F}_{q^r})=\sum_i(-1)^i\,\mathrm{Tr}\Big(\mathrm{Frob}_q^r\mid H^i_{c,\text{ét}}(X_{\overline{\mathbb{F}}_q},\mathbb{Q}_\ell)\Big).$$

Insert this into the exponential defining $Z(X,t)$:

$$\log Z(X,t) = \sum_i (-1)^i \sum_{r\geq 1} \mathrm{Tr}\Big(\mathrm{Frob}_q^r \mid H_c^i\Big) \frac{t^r}{r}.$$

By Chapter 5.1.1,

$$\sum_{r\geq 1} \mathrm{Tr}(\mathrm{Frob}_q^r \,|H_c^i) \frac{t^r}{r} = -\log\det(1 - t\,\mathrm{Frob}_q \,|H_c^i).$$

Therefore

$$Z(X,t) = \prod_i \det\Big(1 - t\,\mathrm{Frob}_q \mid H^i_{c,\text{ét}}(X_{\overline{\mathbb{F}}_q}, \mathbb{Q}_\ell)\Big)^{(-1)^{i+1}}.$$

This is the arithmetic determinant formula. Counting points over all finite extensions is equivalent to taking alternating determinants of Frobenius on compactly supported cohomology. For smooth proper varieties, compactly supported cohomology agrees with ordinary étale cohomology.

5.3 Example: A Smooth Projective Curve

Let $C/\mathbb{F}_q$ be a smooth projective geometrically connected curve of genus g. Its étale cohomology has three non-zero degrees:

$$H^0 \simeq \mathbb{Q}_\ell, \qquad \dim H^1 = 2g, \qquad H^2 \simeq \mathbb{Q}_\ell(-1).$$

Frobenius acts on H^0 by 1, and on H^2 by q. Hence

$$Z(C,t) = \frac{\det(1 - t\,\mathrm{Frob}_q \,|H^1)}{(1-t)(1-qt)}.$$

Set

$$Z(C,t) = \frac{P_C(t)}{(1-t)(1-qt)}, \qquad P_C(t) = \det(1 - t\,\mathrm{Frob}_q \,|H^1).$$

This is the reciprocal characteristic polynomial of Frobenius on H^1: its roots in the t-plane have absolute value $q^{-1/2}$, while the Frobenius eigenvalues have absolute value $q^{1/2}$. The numerator is a determinant on middle cohomology. The denominator comes from the two Tate lines H^0 and H^2.

5.4 Example: An Elliptic Curve

Let $E/\mathbb{F}_q$ be an elliptic curve. Then $g = 1$, so H^1 has dimension 2. Let

$$a_q := q + 1 - \#E(\mathbb{F}_q).$$

The Frobenius characteristic polynomial on H^1 is

$$1 - a_q t + qt^2.$$

Therefore

$$Z(E,t) = \frac{1 - a_q t + qt^2}{(1-t)(1-qt)}.$$

The Riemann hypothesis for $E/\mathbb{F}_q$ says the two reciprocal roots of $1 - a_q t + qt^2$ have absolute value $q^{1/2}$. In determinant language, it is a purity statement about Frobenius eigenvalues on H^1.

5.5 LOCAL L-FACTORS

Let V be an ℓ-adic Galois representation of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$. At a prime $p \neq \ell$, let I_p be inertia. The unramified part is V^{I_p}. The local L-factor is

$$L_p(V,s) = \det\Big(1 - p^{-s}\,\mathrm{Frob}_p \mid V^{I_p}\Big)^{-1}.$$

Again the determinant is the action of Frobenius on a volume line. The Euler product

$$L(V,s) = \prod_p L_p(V,s)$$

is a product of inverse determinants.

5.6 DENINGER-TYPE DETERMINANTS

For $X/\mathbb{F}_q$, Frobenius is a genuine operator on genuine cohomology, and the zeta function is an alternating product of finite determinants.

For arithmetic schemes over $\mathrm{Spec}\,\mathbb{Z}$, Deninger's work asks for an analogue:

$$\zeta_X(s) = \prod_i \det_\infty\Big(s\,\mathrm{id} - \Theta \mid H^i_{\mathrm{arith}}(X)\Big)^{(-1)^{i+1}}.$$

Here $H^i_{\mathrm{arith}}(X)$ is a conjectural arithmetic cohomology, Θ is a Frobenius-like infinitesimal operator, and $\det_\infty$ is a regularised determinant.

For $X = \overline{\mathrm{Spec}\,\mathbb{Z}}$, the expected shape is:

$$\xi_{\mathbb{R}}(s) = \det_\infty\Big(s\,\mathrm{id} - \Theta \mid H^1_{\mathrm{arith}}(\overline{\mathrm{Spec}\,\mathbb{Z}})_{\mathrm{prim}}\Big),$$

while the cohomological degrees 0 and 2 account for the Tate pole lines at $s = 0$ and $s = 1$.

This is Deninger's conjectural determinant template for the arithmetic curve. The open content is the construction of the cohomology, the operator, and the regularised determinant.

5.7 ARAKELOV DETERMINANTS

Arithmetic geometry also has determinant lines with metrics. Let $\pi : X \to \mathrm{Spec}\,\mathbb{Z}$ be an arithmetic surface and let E be a hermitian vector bundle on X. The determinant of cohomology is

$$\lambda(E) = \det R\pi_* E.$$

At the archimedean fibre, analytic torsion supplies a metric on this line, the Quillen metric. The determinant line is therefore Arakelov-geometric: finite primes contribute algebraic cohomology, and the infinite place contributes spectral analysis of Laplacians. Both are determinant constructions.

5.8 FROBENIUS, REGULARISATION, AND ALTERNATING SIGNS

The determinant in arithmetic geometry packages three operations:

(i) finite-place Frobenius acting on cohomology;
(ii) archimedean spectral regularisation;
(iii) alternating cohomological signs.

The determinant is the mechanism by which a cohomological operator becomes a zeta or L-function.

Chapter 6

Geometric Representation Theory

6.1 The Weyl Denominator

Let G be a connected reductive group with maximal torus T, Weyl group W, and positive roots Φ^+. Define

$$\rho := \frac{1}{2} \sum_{\alpha \in \Phi^+} \alpha.$$

The Weyl denominator identity is

$$\sum_{w \in W} (-1)^{\ell(w)} e^{w\rho} = e^{\rho} \prod_{\alpha \in \Phi^+} (1 - e^{-\alpha}).$$

The right side is a determinant. It is the determinant of

$$1 - \mathrm{Ad}(t)^{-1}$$

on the positive nilpotent tangent directions:

$$\prod_{\alpha \in \Phi^+} (1 - e^{-\alpha}).$$

The sign $(-1)^{\ell(w)}$ is the determinant of the Weyl group element acting on the real span of roots.

6.2 The GL_n Model

For $G = GL_n$, the Weyl group is S_n. Let

$$x_i = e^{\varepsilon_i}.$$

The Vandermonde determinant is

$$\det(x_j^{n-i})_{1 \le i, j \le n} = \prod_{i<j} (x_i - x_j).$$

The left side is a determinant of a matrix of weights. The right side is a product over positive roots:

$$\prod_{i<j} x_i(1 - x_j/x_i) = e^{\rho} \prod_{\alpha > 0} (1 - e^{-\alpha})$$

after the standard SL_n normalization. Thus $\prod_{\alpha>0}(1 - e^{-\alpha})$ is the determinant $\det(1 - \mathrm{Ad}(t)^{-1}|\mathfrak{n}_+)$, while e^{ρ} supplies the Weyl half-density. For GL_n the central character accounts for the extra monomial factor.

6.3 Weyl Character Formula

For a dominant weight λ, the Weyl character formula says

$$\operatorname{ch} V_\lambda = \frac{\sum_{w\in W}(-1)^{\ell(w)} e^{w(\lambda+\rho)}}{\sum_{w\in W}(-1)^{\ell(w)} e^{w\rho}}.$$

This is a ratio of two alternating determinants. The denominator is the determinant of the tangent weights of the flag variety at a fixed point. The numerator is the same determinant twisted by the line bundle of weight λ.

The Weyl character formula is the fixed-point formula for equivariant localization on G/B: localization divides by equivariant Euler classes, and these Euler classes are determinants of tangent representations.

6.4 Localization and Euler Classes

Let a torus T act on a smooth proper variety X. At an isolated fixed point $x \in X^T$, the tangent space decomposes into weight spaces:

$$T_x X = \bigoplus_j k_{\chi_j}.$$

The equivariant Euler class is

$$e_T(T_x X) = \prod_j \chi_j.$$

This is the determinant of the T-representation on the tangent space, read multiplicatively in equivariant cohomology.

For $\alpha \in H_T^\bullet(X)$, the Atiyah–Bott localization formula has the form, after inverting the relevant characters,

$$\int_X \alpha = \sum_{x\in X^T} \frac{\alpha|_x}{e_T(T_x X)}.$$

The denominator is a determinant. Geometric representation theory is full of such denominators because representation-theoretic characters are computed by localizing sheaves on spaces with group actions. For non-isolated fixed loci, the sum ranges over components $F \subset X^T$ and the denominator is $e_T(N_F)$.

6.5 Determinant Lines for Tate Vector Spaces

Loop groups force infinite-dimensional linear algebra. Let V be a Tate vector space, such as $k((t))$, and let $L, L' \subset V$ be lattices. Their relative determinant line is

$$\det(L|L') := \det\big(L/(L\cap L')\big) \otimes \det\big(L'/(L\cap L')\big)^\vee.$$

This is finite-dimensional because the two quotient spaces are finite-dimensional when L and L' are commensurable.

The relative determinant line measures the failure of two infinite lattices to have the same volume. If g is a loop acting on V, compare L and gL. The line

$$\det(L|gL)$$

is the determinant anomaly of g. The central extension is the $\mathbb{G}_m$-torsor of nonzero vectors in $\det(L|gL)$; multiplication is induced by the transitivity isomorphism

$$\det(L|gL) \otimes \det(gL|ghL) \simeq \det(L|ghL).$$

6.6 Affine Grassmannian and Kac-Moody Level

For a reductive group G, the affine Grassmannian is

$$\mathrm{Gr}_G = G((t))/G[[t]].$$

Line bundles on Gr_G encode levels of affine Kac-Moody representations. The basic line bundle is a determinant line. For $G = GL_n$, a point of Gr_{GL_n} is a lattice $L \subset k((t))^n$. Relative to the standard lattice

$$L_0 = k[[t]]^n,$$

the determinant line at L is

$$\det(L|L_0).$$

This line is the geometric source of the basic central extension of the loop group. The representation-theoretic level is a determinant class.

6.7 Convolution and Orientation Lines

Geometric representation theory uses correspondences:

$$X \xleftarrow{p} Z \xrightarrow{q} Y.$$

A sheaf on X is pulled back to Z, then pushed forward to Y. When the theory is refined to D-modules, constructible sheaves, or coherent sheaves, the correct functor often carries shifts and orientation lines. For smooth, lci, or derived correspondences these orientation lines are determinant lines of virtual tangent or cotangent complexes:

$$\det\bigl(T_Z^{\mathrm{vir}} - p^*T_X^{\mathrm{vir}} - q^*T_Y^{\mathrm{vir}}\bigr)$$

in Picard-group notation, together with the cohomological rank shift and the chosen Gysin orientation data.

Thus the determinant controls the normalization of convolution. Wrong determinant lines give wrong signs, wrong shifts, and wrong dualities.

6.8 Springer Theory

The Springer resolution

$$\widetilde{\mathcal{N}} = T^*(G/B) \to \mathcal{N}$$

is one of the central spaces of geometric representation theory. The cotangent bundle has a canonical symplectic form. Its canonical bundle is trivial because

$$\omega_{T^*X} \simeq \mathcal{O}_{T^*X}$$

for smooth X. This triviality is a determinant statement: the determinant of the cotangent directions cancels the determinant of the tangent directions. Such cancellations are the geometric reason Springer theory has clean duality and Weyl group actions.

6.9 Euler, Central-Extension, and Orientation Determinants

In geometric representation theory, determinants appear as:

(i) Weyl denominators and character formula denominators;
(ii) equivariant Euler classes in localization;
(iii) determinant line bundles on affine Grassmannians;
(iv) central extensions of loop groups;
(v) orientation lines in convolution;
(vi) duality normalizations in symplectic resolutions.

Every item is the same operation: take a linearized tangent, cotangent, lattice, or cohomology object, pass to its determinant line, and record how symmetry acts on that line.

Chapter 7

The Unified Dictionary

7.1 Action on the Volume Line

The determinant is the passage from a linear operator on a space to the induced scalar action on the space's volume line.

7.2 Determinant Outputs Across Fields

Field	Space	Determinant output
Linear algebra	finite vector space V	scalar $\det(T)$ on $\det V$
Algebraic geometry	vector bundle or perfect complex	determinant line bundle
Arithmetic geometry	cohomology with Frobenius	zeta and L-factors
Geometric representation theory	tangent, lattice, or sheaf-theoretic data	Euler class, denominator, central extension, level

7.3 Determinant Data

When a claimed determinant appears, test it by three questions.

(i) What is the underlying linear or derived object?
(ii) What is its determinant line?
(iii) Which operator, symmetry, or Frobenius acts on that line?

Naming these three objects is the construction of the determinant; without them the displayed symbol is only notation.

7.4 Number-Field Determinant Data

For a finite-dimensional operator, the determinant is automatic. An infinite-dimensional arithmetic operator requires extra data. A formula

$$\xi(s) = \det_{\infty}(s\,\mathrm{id} - \Theta)$$

requires:

(i) a cohomological space;
(ii) a densely-defined operator Θ;
(iii) a spectral theory for Θ;
(iv) a regularised determinant;
(v) a trace formula linking the determinant to primes and archimedean terms.

The finite-field formula gives the template. The number-field object requires a separate construction, which is the substance of a Deninger-type arithmetic cohomology.

Chapter 8

The Virtual K_0-Determinant on $K3 \times E$

The preceding chapters realised the determinant as a finite-dimensional exterior-power scalar, a determinant line bundle on a perfect complex, a Fredholm or zeta-regularised eigenvalue product, an arithmetic L-factor, and a Weyl-type denominator. One realisation remains: a virtual K_0-determinant whose value on a Calabi–Yau threefold is the Igusa cusp form Δ_5, the second-quantised denominator of the Mukai-K3 Borcherds–Kac–Moody algebra $\mathfrak{g}_{\Delta_5}$. This chapter constructs that realisation, names its hypothesis package, and isolates the operator-level Pfaffian as a separate construction.

8.1 Type signature

The construction carries the type signature

$$(\text{quadrant} = \mathrm{CY}_3, \quad \text{presentation} = \text{Borcherds product}, \quad \text{level} = 5, \quad \text{hypothesis package} = \text{Definition 8.1.1}).$$

Level 5 in the Beilinson tower is the modular-shadow level: a κ-tuple, partition function, or modular form. The construction is therefore a *scalar shadow*, not an operator algebra. The distinction is enforced by the shadow-object distinction1 (§8.3); the operator-level Pfaffian is the open frontier F11 (§8.4).

Definition 8.1.1 *(hypothesis package for the virtual K_0-determinant).* The virtual K_0-determinant on $K3 \times E$ is constructed under the package:

(i) **Mukai stack and Heegner reciprocity.** $\mathcal{M}_{K3}$ is the derived moduli stack of objects in $D^b\,\mathrm{Coh}(K3)$ with bounded Mukai vector, and $K_0(\mathcal{M}_{K3})$ is the Grothendieck group of perfect complexes. The Bruinier–Heegner Chern-class reciprocity supplies a natural map $K_0(\mathcal{M}_{K3}) \to H^*(\mathcal{M}_{K3}, \mathbb{Z})$ whose pairing with the Heegner divisors of [3] matches Fourier coefficients of $\phi_{0,1}$ at signed Heegner indices.

(ii) **Reduction to $K3 \times E$.** The structure-group reduction picks out the locus where the K3 factor carries a hyperkähler structure and the elliptic factor carries a complex structure compatible with a Lagrangian $E \subset \mathcal{M}_{K3}$. The resulting structure group is the paramodular group $\mathbf{Sp}_4(\mathbb{Z})$ acting on $\mathbb{H}_2$, the Siegel upper half space.

(iii) **Borcherds singular-theta lift of $\phi_{0,1}$.** The weak Jacobi form

$$\phi_{0,1}(\tau, z) = \tfrac{1}{2} Z_{\mathrm{K3}}(\tau, z) = \sum_{n \geq 0,\, l \in \mathbb{Z}} f(n, l)\, q^n r^l$$

has constant term $c_1(0) = f(0,0) + 2f(0,1) + \cdots = 10$, and feeds the Borcherds singular-theta lift to give a holomorphic modular form of weight 5 with respect to the multiplier system χ_5 of $\mathbf{Sp}_4(\mathbb{Z})$. This is Δ_5.

(iv) **Virtual super vector space data.** For each $(n, l) \in \mathbb{N}_{\geq 0} \times \mathbb{Z}$, choose a virtual super vector space $\mathbb{U}_{n,l}$ with $\mathrm{sdim}\, \mathbb{U}_{n,l} = f(n, l)$. Set $\mathcal{V}_{(n,l,m)} := \mathbb{U}_{nm,l}$.

(v) **Effective semigroup.** $\Gamma_{\mathrm{eff}} \subset \mathbb{Z}^3$ is the saturated cone generated by the support of $\phi_{0,1}$ on the standard basis $(2f_2, -f_3, 2f_{-2})$.

8.2 The virtual K_0-determinant

Definition 8.2.1 *(virtual K_0-determinant).* With the data of Definition 8.1.1, the virtual K_0-determinant is the formal symmetric power

$$\mathcal{D}_X(Z) := 64\, q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} \mathrm{sdet}\big(1 - q^n r^l s^m\,;\, \mathcal{V}_{(n,l,m)}\big),$$

where $\mathrm{sdet}(1-t\,;\,V) := (1-t)^{\mathrm{sdim}\,V}$ extends the finite-dimensional determinant of $\mathrm{id} - tT$ on a virtual super vector space, with multiplicative continuation in the variable t. The constant 64 is the theta-constant normalisation $[q^{1/2} r^{1/2} s^{1/2}]\Delta_5 = 64$.

Theorem 8.2.2 *(K_0-determinant equals the Igusa cusp form).* With the data of Definition 8.1.1 and the convention of Definition 8.2.1,

$$\mathcal{D}_X(Z) = 64\, q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)} = \Delta_5(Z).$$

The monic Borcherds product is $D_5 = 64^{-1}\Delta_5$, satisfying $\Phi_{10}^{\mathrm{OP}} = D_5^2 = 4096^{-1}\Delta_5^2$ in Oberdieck–Pandharipande normalisation.

Proof. Apply the determinant identity $\mathrm{sdet}(1-t\,;\,V) = \exp\big(-\sum_{r\geq 1} \frac{t^r}{r}\,\mathrm{sdim}\,V\big)$ to each factor; the multiplicative form is forced by $\mathrm{sdim}\,\mathbb{U}_{nm,l} = f(nm,l)$. The resulting product expansion is the Borcherds singular-theta lift of $\phi_{0,1}$ to $\mathbb{H}_2$, evaluated at the cusp where the multiplier system is trivial; by [1, 2] this lift is Δ_5 up to the normalisation 64. The conversion $\Phi_{10}^{\mathrm{OP}} = 4096^{-1}\Delta_5^2$ is the squared monic form, with $4096 = 64^2$. □

Proposition 8.2.3 *(universal Borcherds-weight identity at $N = 1$).* The Borcherds weight of Δ_5 is

$$\kappa_{\mathrm{BKM}}(\Delta_5) \;=\; \tfrac{1}{2}\, c_1(0) \;=\; 5.$$

Proof. Three independent paths.

Path 1 (Borcherds singular-theta lift). The lift carries weak Jacobi forms of weight 0 to Siegel modular forms of weight $c(0)/2$, where $c(0)$ is the constant Fourier coefficient [1, Thm. 13.3]. For $2\phi_{0,1}$, the constant term is $c_1(0) = 2\cdot 5 = 10$, giving weight 5.

Path 2 (Gritsenko–Nikulin denominator identity). The Borcherds–Kac–Moody superalgebra $\mathfrak{g}_{\Delta_5}$ has denominator identity $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ [2, §2]. The denominator function transforms under $\mathbf{Sp}_4(\mathbb{Z})$ with character χ_5 of weight 5, fixed by the Weyl-vector formula $\rho = \frac{1}{2}(2f_2 - f_3 + 2f_{-2})$ and the imaginary-root multiplicities $\mathrm{smult}\,\mathfrak{g}_{\Delta_5,\alpha(n,l,m)} = f(nm,l)$.

Path 3 (Siegel modular form weight). As an element of $M_5(\mathbf{Sp}_4(\mathbb{Z}),\chi_5)$, the Igusa cusp form Δ_5 has weight 5 by direct construction as a Maass lift of $\phi_{0,1}$ [4, 5]. Independent verification: the Fourier expansion at the standard cusp begins $\Delta_5 = \chi_5 \cdot q^{1/2} r^{1/2} s^{1/2}(1 + O(q,r,s))$ and the q-expansion at the Maass-lift cusp matches the universal Saito–Kurokawa input weight 4 shifted by +1 to weight 5 via the $\chi_{10} = \Delta_5^2$ identification with the Igusa cusp form of weight 10 [5, §6].

The three paths agree: $\kappa_{\mathrm{BKM}}(\Delta_5) = 5$. The identity is the $N = 1$ instance of the universal Borcherds-weight identity $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ for $N \in \{1,2,3,4,6\}$ plus half- and quarter-integer continuations [7, Vol. III, FRONTIER §"Universal Borcherds-weight identity"]. □

Remark 8.2.4 *(Borcherds weight versus chiral conductor).* The K3×E row of the five-archetype 5×5 κ-stratification matrix carries

$$(\kappa_{\mathrm{cat}}, \kappa_{\mathrm{ch}}^{\mathrm{Hodge}}, \kappa_{\mathrm{ch}}^{\mathrm{Heis}}, \kappa_{\mathrm{BKM}}, \kappa_{\mathrm{fiber}}) = (0, 0, 3, 5, 24).$$

The five entries are five distinct κ-measurements, not one invariant in five guises (master pattern five-invariant). The Mukai-K3 conductor $K_{\mathcal{B}}^{\kappa_{\mathrm{ch}}} = 2c_+(\mathrm{Mukai}(K3)) = 8$ is yet a sixth scalar, again on a separate lane: it is the Mukai positive-signature half, not the Borcherds weight. The additive formula $\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}} + \chi(\mathcal{O}_{\mathrm{fiber}})$ fails at every N, including the $N = 1$ K3×E row where it would give $0 + 0 = 0$ versus the correct 5.

8.3 Scalar shadow versus operator algebra

The deepest false pattern in this domain is the identification of a scalar shadow with the operator algebra it shadows (master pattern scalar-operator). The Igusa cusp form Δ_5 is a level-5 scalar; it is not a level-4 operator algebra.

Proposition 8.3.1 *(scalar shadow versus operator algebra).* The Igusa cusp form $\Delta_5 \in M_5(\mathbf{Sp}_4(\mathbb{Z}), \chi_5)$ is a Siegel modular form. As a scalar at level 5 of the Beilinson tower it is the modular trace and clutching of an operator algebra $\mathfrak{g}_{\Delta_5}$ at level 4. The reverse direction

$$\Delta_5 \rightsquigarrow \mathfrak{g}_{\Delta_5}$$

is not a determinant identity. It requires an operator-level Pfaffian / square-root reconstruction whose existence on a compact $K3 \times E$ source is an open problem.

Proof. The forward direction is the Borcherds–Gritsenko–Nikulin denominator identity $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$: the BKM superalgebra produces the Siegel cusp form via Weyl-character data [1, 2]. The scalar reads off the operator-graded data; this direction is well-defined.

The reverse direction proposes to recover $\mathfrak{g}_{\Delta_5}$ as the operator algebra of a compact $K3 \times E$ source whose protected trace is Δ_5. By the exterior-power machinery of Chapter 1, a determinant identity does not invert: the top exterior line records eigenvalue products, not eigenvectors. To recover the operator algebra one must solve, in addition,

(i) the orientation/Pfaffian datum that distinguishes Δ_5 from $-\Delta_5$ (the squared scalar $-4096\,\Delta_5^{-2} = Z^X_{\mathrm{OP/DT}}$ is orientation-forgetful);
(ii) the Hall bracket and Hopf pairing that promote $\mathfrak{g}_{\Delta_5}$ to a compact Hall source;
(iii) primitive root representatives, the radical quotient, the no-extra-relations theorem, PBW, and parity dimensions on every finite root window.

None of these is a determinant. The forward direction is the determinant theorem; the reverse direction is the open problem F11 of the Vol I frontier. □

Corollary 8.3.2 *(Scalar shadow and operator algebra).* Δ_5 is the Borcherds denominator at level 5. The level-4 operator algebra is reconstructed only from operator data; the scalar denominator supplies its shadow and not its Hilbert space, path integral, or Pfaffian algebra.

8.4 FII: the operator-level Pfaffian as open frontier

The virtual K_0-determinant package *produces* the scalar Δ_5. It does *not* produce a compact protected operator $\mathfrak{D}_X$ on $K3 \times E$ whose protected Pfaffian is Δ_5.

Problem 8.4.1 *(FII — operator-level Pfaffian on $K3 \times E$).* Construct a first-order protected operator/algebra $\mathfrak{D}_X$ on a compact $K3 \times E$ source, and a comparison map between the Pfaffian line of $\mathfrak{D}_X$ and the automorphic line of Δ_5, such that

$$\mathrm{Pf}(\mathfrak{D}_X) \;=\; \Delta_5, \qquad \mathrm{Tr}_{\mathrm{prot}}\big((-1)^F \mathfrak{D}_X^{-2}\big) \;=\; -4096\,\Delta_5^{-2} \;=\; Z^X_{\mathrm{OP/DT}}.$$

The orientation/Pfaffian datum must distinguish Δ_5 from $-\Delta_5$ (squared scalar is orientation-forgetful) and the construction must include:

- (i) additive formal dyonic charge lattice Γ_X^{form} with Gram map Π_X to Γ_{gram};
- (ii) holomorphic E_3-factorisation algebra on $K3 \times E$;
- (iii) normal-ordered extension $\widehat{\Gamma}_X = \Gamma_X^{\mathrm{form}} \oplus_B \Gamma_{\mathrm{gram}}$ rectifying the failure of strict fixed-lift raw Gram descent for the type-II real-root brackets;
- (iv) formality datum, QME, anomaly control, descent, and finite-first HN completion;
- (v) chiral specialisation along the K3 direction to a chiral object on $\mathrm{Ran}(E)$, with the $\mathrm{Ran}(E)$-support-locality obstruction solved by mixed local/wrapped correspondences.

Remark 8.4.2 *(Compact-realisation reduction).* The compact-realisation problem factors through the local Borcherds and Hall–Drinfeld target data of [6] and the CY-to-chiral two-stage functor of [7], $\Phi_3^{(\Sigma_2,E)} = \mathrm{Sp}^{\mathrm{ch}}_{\Sigma_2,E} \circ \Phi_3^{\mathrm{FA}}$ with the K3 chiral bialgebra $\mathbf{H}_{\Delta_5}$ as Stage 2 output. The compact realisation problem is finite-stage in the sense of [6, §Compact realization]: construct geometric Hall–Dirac source data and a compatible functor to the algebraic target $G(\mathscr{B}_{\Delta_5})_W$ on every downward-saturated finite root window W. Each step has a named obstruction; none is automatic from the scalar.

Remark 8.4.3 *(diagnostic: where the determinant chain ends).* The determinant chain of this paper extends as follows.

level 0	exterior power $\bigwedge^n V$
level 1	determinant line bundle $\det E$ on a perfect complex
level 2	determinant of cohomology $\det R\Gamma(X, E)$
level 3	Knudsen–Mumford determinant of a complex
level 4	Quillen / zeta / Fredholm regularised determinant
level 5	virtual K_0-determinant $\mathcal{D}_X = \Delta_5$ on the Mukai stack of K3
level 6	operator-level Pfaffian $\mathrm{Pf}(\mathfrak{D}_X) = \Delta_5$ (FII, open)

The determinant construction closes at level 5. Level 6 is the second-quantised string construction, distinct in level and in hypothesis package; it is not produced by the determinant chain running upward without further data.

8.5 Verification: $\varkappa_{\mathrm{BKM}}(\Delta_5) = 5$ by three paths

The numerical claim $\varkappa_{\mathrm{BKM}}(\Delta_5) = 5$ is verified by three independent paths in Proposition 8.2.3: the Borcherds singular-theta lift constant-term formula $c(0)/2 = 5$, the Gritsenko–Nikulin denominator identity weight, and the direct Saito–Kurokawa lift weight of Δ_5 as a Siegel modular form.

A fourth, dimensional path: the universal Borcherds-weight identity $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ for $N \in \{1, 2, 3, 4, 6\}$ at $N = 1$ (paramodular K3×E), with constant term $c_1(0) = 10$, gives weight 5. The same identity at $N = 12$ (Fake Monster on $\mathrm{II}_{25,1}$) gives weight 12, with input denominator $\theta_{\Lambda_{24}}/\eta^{24}$ and constant term 24. The two rows are not re-indexings of each other; the Cartans $\Lambda_{\mathrm{II}}^{2,1}$ and $\mathrm{II}_{25,1}$ admit no signature-preserving embedding, and the Fourier channels at the two weights do not interfere. The identity therefore holds row by row, with the input denominator declared, and is verified at $N = 1$ for Δ_5 to give weight 5.

8.6 Cross-volume placement

The virtual K_0-determinant package sits at level 5 of the Vol I Open-quadrant Beilinson tower, paired by holographic vertical equivalence with the Vol III CY-quadrant statement $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ at $N = 1$. The Vol III two-stage functor $\Phi_3^{(\Sigma_2,E)} = \mathrm{Sp}^{\mathrm{ch}}_{\Sigma_2,E} \circ \Phi_3^{\mathrm{FA}}$ produces, from the Calabi–Yau category $D^b\,\mathrm{Coh}(K3 \times E)$, the K3 chiral bialgebra $\mathbf{H}_{\Delta_5}$ with $\mathrm{ChirHoch}^1(\mathbf{H}_{\Delta_5}) \cong \mathfrak{g}_{\Delta_5}/\mathfrak{h}$, the BKM superalgebra modulo its imaginary Cartan [7]. The scalar Δ_5 constructed here is the denominator of that BKM superalgebra; the chiral bialgebra $\mathbf{H}_{\Delta_5}$ is its level-4 operator companion (modulo the F11 compact-realisation problem).

8.7 Summary

The determinant chain extends into the modular-shadow regime. The virtual K_0-determinant package has scalar value Δ_5 on $K3 \times E$, a Siegel modular form of weight 5. The Borcherds weight identity $\kappa_{\mathrm{BKM}}(\Delta_5) = c_1(0)/2 = 5$ holds by three independent paths. The hypothesis package names the Mukai stack, the Bruinier–Heegner Chern-class reciprocity, the structure-group reduction to $K3 \times E$, and the Borcherds singular-theta lift of $\phi_{0,1}$. The scalar-operator discipline is enforced: Δ_5 is the scalar shadow at level 5, not the operator algebra at level 4. The construction of an operator-level Pfaffian whose value is Δ_5 is the open frontier F11.

Conclusion

The determinant answers a precise linear question:

by what scalar does T multiply volume?

The rigorous answer is:

$$\bigwedge^{n} T = \det(T) \cdot \mathrm{id}_{\bigwedge^{n} V} .$$

Algebraic geometry globalizes the volume line into a determinant line bundle. Arithmetic geometry makes Frobenius act on cohomological volume lines and obtains zeta functions. Geometric representation theory makes symmetry act on tangent, lattice, and sheaf-theoretic determinant lines and obtains denominators, levels, Euler classes, and central extensions. The virtual K_0-determinant of the Mukai stack of K3 sheaves, evaluated on $K3 \times E$ with Borcherds singular-theta input $\phi_{0,1}$, gives the Igusa cusp form Δ_5 of weight 5 — a Borcherds–Kac–Moody denominator, the second-quantised string scalar, and the level-5 modular shadow of an operator algebra still to be constructed.

Across finite locally free, perfect, regularised, and virtual K_0 settings, the determinant remains the action on the appropriate volume line:

$$\boxed{\text{determinant} = \text{action on the top exterior line.}}$$

The construction of the operator-level Pfaffian $\mathrm{Pf}(\mathfrak{D}_X) = \Delta_5$ on a compact $K3 \times E$ source is distinct in level and hypothesis package from the determinant of this paper; it is the open frontier F11.

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